

GEOMETRY

(Common Core)

Friday, June 16, 2017 — 9:15 a.m. to 12:15 p.m., only

Student Name: _____

School Name: _____

The possession or use of any communications device is strictly prohibited when taking this examination. If you have or use any communications device, no matter how briefly, your examination will be invalidated and no score will be calculated for you.

Print your name and the name of your school on the lines above.

A separate answer sheet for **Part I** has been provided to you. Follow the instructions from the proctor for completing the student information on your answer sheet.

This examination has four parts, with a total of 36 questions. You must answer all questions in this examination. Record your answers to the Part I multiple-choice questions on the separate answer sheet. Write your answers to the questions in **Parts II, III, and IV** directly in this booklet. All work should be written in pen, except for graphs and drawings, which should be done in pencil. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale.

The formulas that you may need to answer some questions in this examination are found at the end of the examination. This sheet is perforated so you may remove it from this booklet.

Scrap paper is not permitted for any part of this examination, but you may use the blank spaces in this booklet as scrap paper. A perforated sheet of scrap graph paper is provided at the end of this booklet for any question for which graphing may be helpful but is not required. You may remove this sheet from this booklet. Any work done on this sheet of scrap graph paper will *not* be scored.

When you have completed the examination, you must sign the statement printed at the end of the answer sheet, indicating that you had no unlawful knowledge of the questions or answers prior to the examination and that you have neither given nor received assistance in answering any of the questions during the examination. Your answer sheet cannot be accepted if you fail to sign this declaration.

Notice...

A graphing calculator, a straightedge (ruler), and a compass must be available for you to use while taking this examination.

DO NOT OPEN THIS EXAMINATION BOOKLET UNTIL THE SIGNAL IS GIVEN.

High School Math Reference Sheet

1 inch = 2.54 centimeters	1 kilometer = 0.62 mile	1 cup = 8 fluid ounces
1 meter = 39.37 inches	1 pound = 16 ounces	1 pint = 2 cups
1 mile = 5280 feet	1 pound = 0.454 kilogram	1 quart = 2 pints
1 mile = 1760 yards	1 kilogram = 2.2 pounds	1 gallon = 4 quarts
1 mile = 1.609 kilometers	1 ton = 2000 pounds	1 gallon = 3.785 liters
		1 liter = 0.264 gallon
		1 liter = 1000 cubic centimeters

Triangle	$A = \frac{1}{2}bh$
Parallelogram	$A = bh$
Circle	$A = \pi r^2$
Circle	$C = \pi d$ or $C = 2\pi r$
General Prisms	$V = Bh$
Cylinder	$V = \pi r^2 h$
Sphere	$V = \frac{4}{3}\pi r^3$
Cone	$V = \frac{1}{3}\pi r^2 h$
Pyramid	$V = \frac{1}{3}Bh$

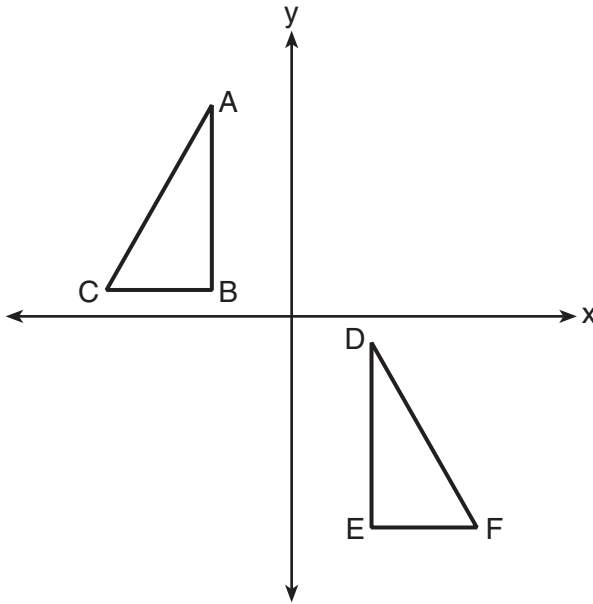
Pythagorean Theorem	$a^2 + b^2 = c^2$
Quadratic Formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Arithmetic Sequence	$a_n = a_1 + (n - 1)d$
Geometric Sequence	$a_n = a_1 r^{n-1}$
Geometric Series	$S_n = \frac{a_1 - a_1 r^n}{1 - r}$ where $r \neq 1$
Radians	1 radian = $\frac{180}{\pi}$ degrees
Degrees	1 degree = $\frac{\pi}{180}$ radians
Exponential Growth/Decay	$A = A_0 e^{k(t-t_0)} + B_0$

Part I

Answer all 24 questions in this part. Each correct answer will receive 2 credits. No partial credit will be allowed. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For each statement or question, choose the word or expression that, of those given, best completes the statement or answers the question. Record your answers on your separate answer sheet. [48]

Use this space for
computations.

- 1 In the diagram below, $\triangle ABC \cong \triangle DEF$.

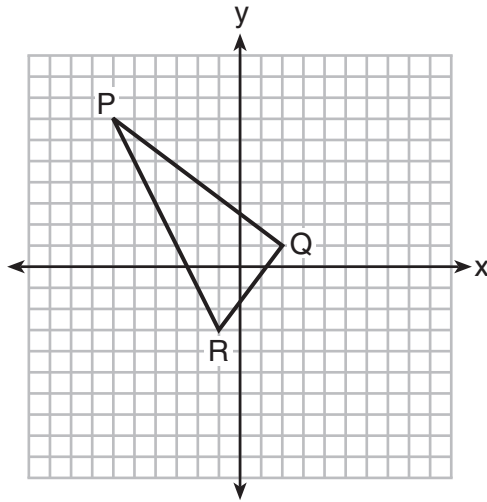


Which sequence of transformations maps $\triangle ABC$ onto $\triangle DEF$?

- (1) a reflection over the x -axis followed by a translation
- (2) a reflection over the y -axis followed by a translation
- (3) a rotation of 180° about the origin followed by a translation
- (4) a counterclockwise rotation of 90° about the origin followed by a translation

**Use this space for
computations.**

- 2** On the set of axes below, the vertices of $\triangle PQR$ have coordinates $P(-6,7)$, $Q(2,1)$, and $R(-1,-3)$.

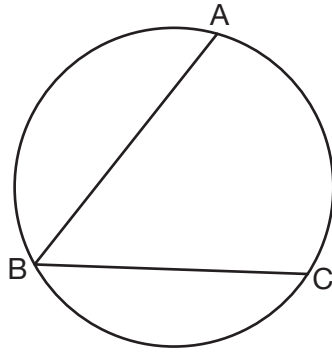


What is the area of $\triangle PQR$?

- | | |
|--------|--------|
| (1) 10 | (3) 25 |
| (2) 20 | (4) 50 |
- 3** In right triangle ABC , $m\angle C = 90^\circ$. If $\cos B = \frac{5}{13}$, which function also equals $\frac{5}{13}$?
- | | |
|--------------|--------------|
| (1) $\tan A$ | (3) $\sin A$ |
| (2) $\tan B$ | (4) $\sin B$ |

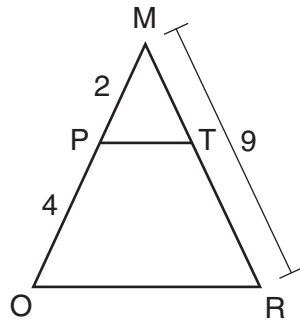
Use this space for
computations.

- 4 In the diagram below, $m\widehat{ABC} = 268^\circ$.



What is the number of degrees in the measure of $\angle ABC$?

- (1) 134° (3) 68°
(2) 92° (4) 46°
- 5 Given $\triangle MRO$ shown below, with trapezoid $PTRO$, $MR = 9$, $MP = 2$, and $PO = 4$.



What is the length of $\overline{TR}$?

- (1) 4.5 (3) 3
(2) 5 (4) 6

**Use this space for
computations.**

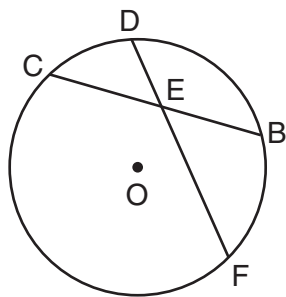
6 A line segment is dilated by a scale factor of 2 centered at a point not on the line segment. Which statement regarding the relationship between the given line segment and its image is true?

- (1) The line segments are perpendicular, and the image is one-half of the length of the given line segment.
- (2) The line segments are perpendicular, and the image is twice the length of the given line segment.
- (3) The line segments are parallel, and the image is twice the length of the given line segment.
- (4) The line segments are parallel, and the image is one-half of the length of the given line segment.

7 Which figure always has exactly four lines of reflection that map the figure onto itself?

- (1) square
- (2) rectangle
- (3) regular octagon
- (4) equilateral triangle

8 In the diagram below of circle O , chord $\overline{DF}$ bisects chord $\overline{BC}$ at E .

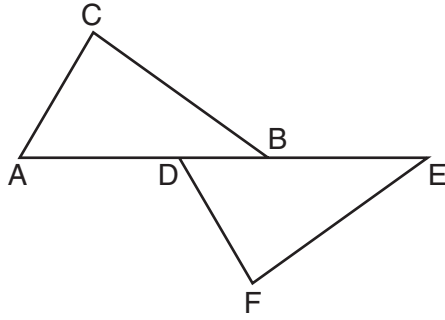


If $BC = 12$ and FE is 5 more than DE , then FE is

- (1) 13
- (2) 9
- (3) 6
- (4) 4

Use this space for
computations.

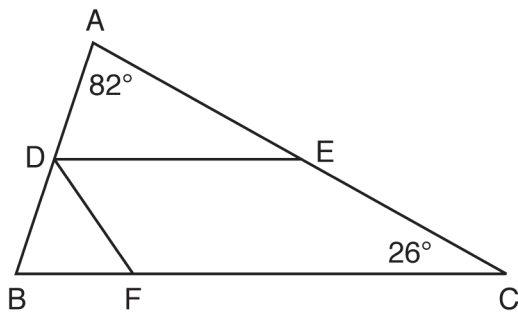
9 Kelly is completing a proof based on the figure below.



She was given that $\angle A \cong \angle EDF$, and has already proven $\overline{AB} \cong \overline{DE}$. Which pair of corresponding parts and triangle congruency method would *not* prove $\triangle ABC \cong \triangle DEF$?

- | | |
|---|---|
| (1) $\overline{AC} \cong \overline{DF}$ and SAS | (3) $\angle C \cong \angle F$ and AAS |
| (2) $\overline{BC} \cong \overline{EF}$ and SAS | (4) $\angle CBA \cong \angle FED$ and ASA |

10 In the diagram below, $\overline{DE}$ divides $\overline{AB}$ and $\overline{AC}$ proportionally, $m\angle C = 26^\circ$, $m\angle A = 82^\circ$, and $\overline{DF}$ bisects $\angle BDE$.



The measure of angle DFB is

- | | |
|----------------|----------------|
| (1) 36° | (3) 72° |
| (2) 54° | (4) 82° |

Use this space for
computations.

11 Which set of statements would describe a parallelogram that can always be classified as a rhombus?

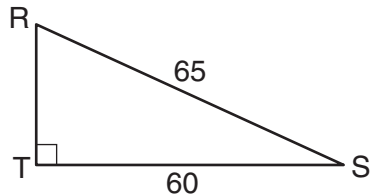
- I. Diagonals are perpendicular bisectors of each other.
- II. Diagonals bisect the angles from which they are drawn.
- III. Diagonals form four congruent isosceles right triangles.

- (1) I and II
- (2) I and III
- (3) II and III
- (4) I, II, and III

12 The equation of a circle is $x^2 + y^2 - 12y + 20 = 0$. What are the coordinates of the center and the length of the radius of the circle?

- (1) center (0,6) and radius 4
- (2) center (0,−6) and radius 4
- (3) center (0,6) and radius 16
- (4) center (0,−6) and radius 16

13 In the diagram of $\triangle RST$ below, $m\angle T = 90^\circ$, $RS = 65$, and $ST = 60$.



What is the measure of $\angle S$, to the *nearest degree*?

- (1) 23°
- (2) 43°
- (3) 47°
- (4) 67°

**Use this space for
computations.**

- 14** Triangle $A'B'C'$ is the image of $\triangle ABC$ after a dilation followed by a translation.

Which statement(s) would always be true with respect to this sequence of transformations?

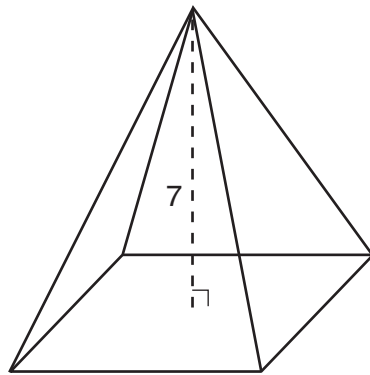
- I. $\triangle ABC \cong \triangle A'B'C'$
- II. $\triangle ABC \sim \triangle A'B'C'$
- III. $\overline{AB} \parallel \overline{A'B'}$
- IV. $AA' = BB'$

- (1) II, only
- (2) I and II
- (3) II and III
- (4) II, III, and IV

- 15** Line segment RW has endpoints $R(-4,5)$ and $W(6,20)$. Point P is on $\overline{RW}$ such that $RP:PW$ is 2:3. What are the coordinates of point P ?

- (1) (2,9)
- (2) (0,11)
- (3) (2,14)
- (4) (10,2)

- 16** The pyramid shown below has a square base, a height of 7, and a volume of 84.

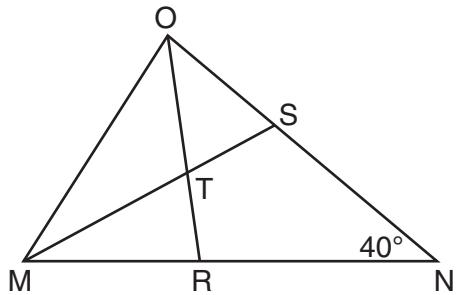


What is the length of the side of the base?

- (1) 6
- (2) 12
- (3) 18
- (4) 36

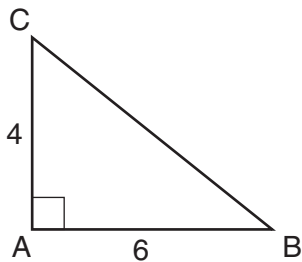
Use this space for
computations.

- 17 In the diagram below of triangle MNO , $\angle M$ and $\angle O$ are bisected by $\overline{MS}$ and $\overline{OR}$, respectively. Segments MS and OR intersect at T , and $m\angle N = 40^\circ$.



If $m\angle TMR = 28^\circ$, the measure of angle OTS is

- (1) 40° (3) 60°
(2) 50° (4) 70°
- 18 In the diagram below, right triangle ABC has legs whose lengths are 4 and 6.



What is the volume of the three-dimensional object formed by continuously rotating the right triangle around $\overline{AB}$?

- (1) 32π (3) 96π
(2) 48π (4) 144π

Use this space for
computations.

19 What is an equation of a line that is perpendicular to the line whose equation is $2y = 3x - 10$ and passes through $(-6, 1)$?

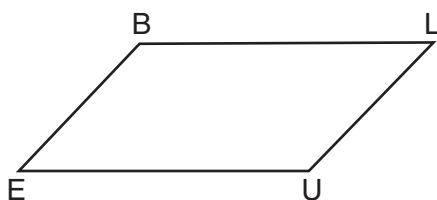
(1) $y = -\frac{2}{3}x - 5$

(3) $y = \frac{2}{3}x + 1$

(2) $y = -\frac{2}{3}x - 3$

(4) $y = \frac{2}{3}x + 10$

20 In quadrilateral $BLUE$ shown below, $\overline{BE} \cong \overline{UL}$.



Which information would be sufficient to prove quadrilateral $BLUE$ is a parallelogram?

(1) $\overline{BL} \parallel \overline{EU}$

(3) $\overline{BE} \cong \overline{BL}$

(2) $\overline{LU} \parallel \overline{BE}$

(4) $\overline{LU} \cong \overline{EU}$

21 A ladder 20 feet long leans against a building, forming an angle of 71° with the level ground. To the *nearest foot*, how high up the wall of the building does the ladder touch the building?

(1) 15

(3) 18

(2) 16

(4) 19

22 In the two distinct acute triangles ABC and DEF , $\angle B \cong \angle E$. Triangles ABC and DEF are congruent when there is a sequence of rigid motions that maps

(1) $\angle A$ onto $\angle D$, and $\angle C$ onto $\angle F$

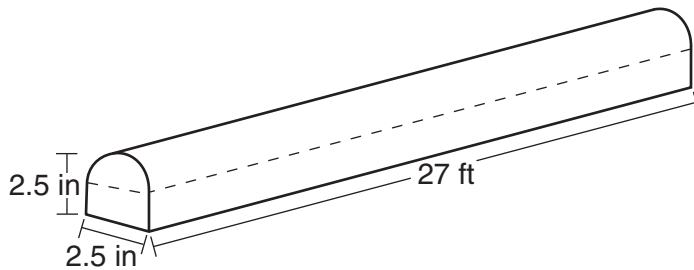
(2) $\overline{AC}$ onto $\overline{DF}$, and $\overline{BC}$ onto $\overline{EF}$

(3) $\angle C$ onto $\angle F$, and $\overline{BC}$ onto $\overline{EF}$

(4) point A onto point D , and $\overline{AB}$ onto $\overline{DE}$

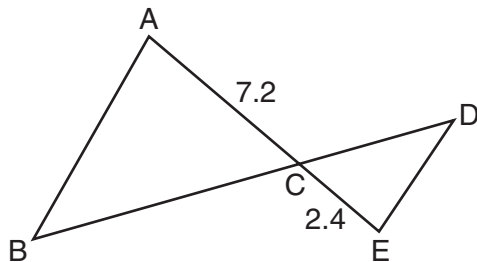
Use this space for
computations.

- 23 A fabricator is hired to make a 27-foot-long solid metal railing for the stairs at the local library. The railing is modeled by the diagram below. The railing is 2.5 inches high and 2.5 inches wide and is comprised of a rectangular prism and a half-cylinder.



How much metal, to the *nearest cubic inch*, will the railing contain?

- (1) 151 (3) 1808
(2) 795 (4) 2025
- 24 In the diagram below, $AC = 7.2$ and $CE = 2.4$.



Which statement is *not* sufficient to prove $\triangle ABC \sim \triangle EDC$?

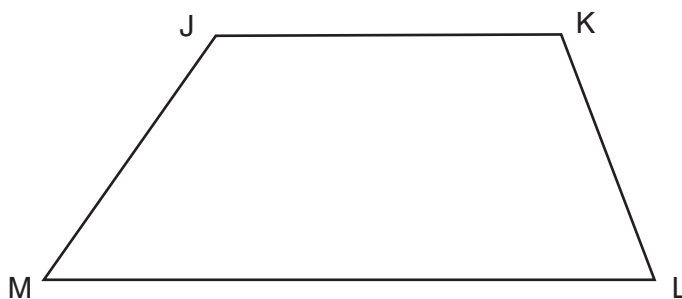
- (1) $\overline{AB} \parallel \overline{ED}$
(2) $DE = 2.7$ and $AB = 8.1$
(3) $CD = 3.6$ and $BC = 10.8$
(4) $DE = 3.0$, $AB = 9.0$, $CD = 2.9$, and $BC = 8.7$
-

Part II

Answer all 7 questions in this part. Each correct answer will receive 2 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [14]

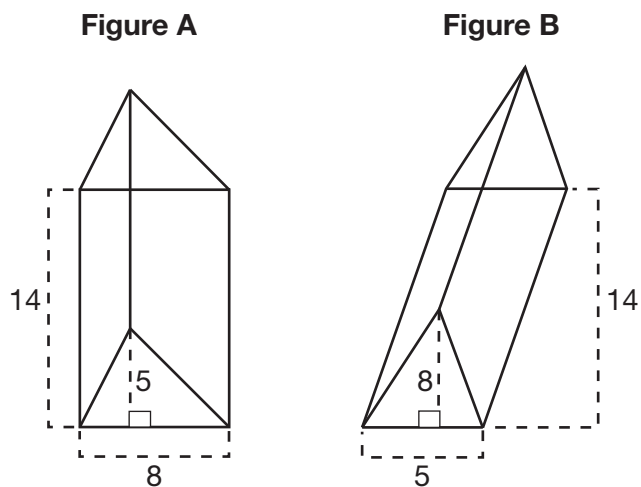
25 Given: Trapezoid $JKLM$ with $\overline{JK} \parallel \overline{ML}$

Using a compass and straightedge, construct the altitude from vertex J to $\overline{ML}$.
[Leave all construction marks.]



26 Determine and state, in terms of π , the area of a sector that intercepts a 40° arc of a circle with a radius of 4.5.

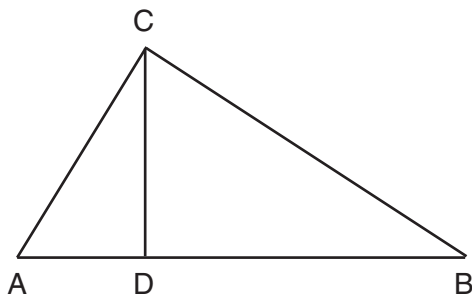
- 27** The diagram below shows two figures. Figure *A* is a right triangular prism and figure *B* is an oblique triangular prism. The base of figure *A* has a height of 5 and a length of 8 and the height of prism *A* is 14. The base of figure *B* has a height of 8 and a length of 5 and the height of prism *B* is 14.



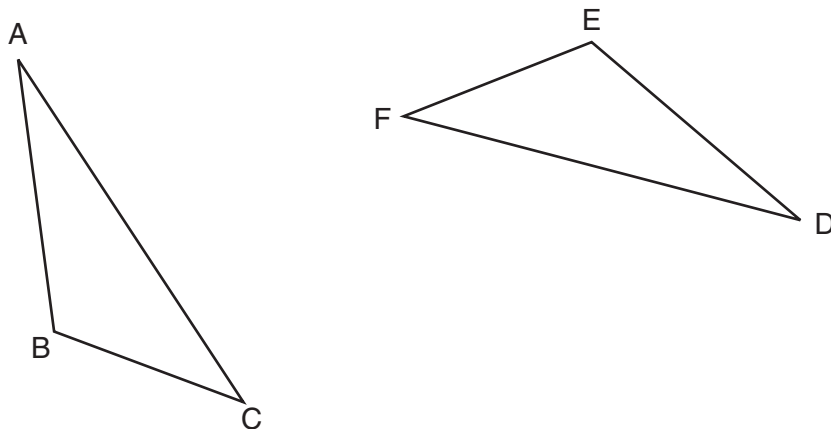
Use Cavalieri's Principle to explain why the volumes of these two triangular prisms are equal.

- 28** When volleyballs are purchased, they are not fully inflated. A partially inflated volleyball can be modeled by a sphere whose volume is approximately 180 in^3 . After being fully inflated, its volume is approximately 294 in^3 . To the *nearest tenth of an inch*, how much does the radius increase when the volleyball is fully inflated?

- 29** In right triangle ABC shown below, altitude $\overline{CD}$ is drawn to hypotenuse $\overline{AB}$. Explain why $\triangle ABC \sim \triangle ACD$.



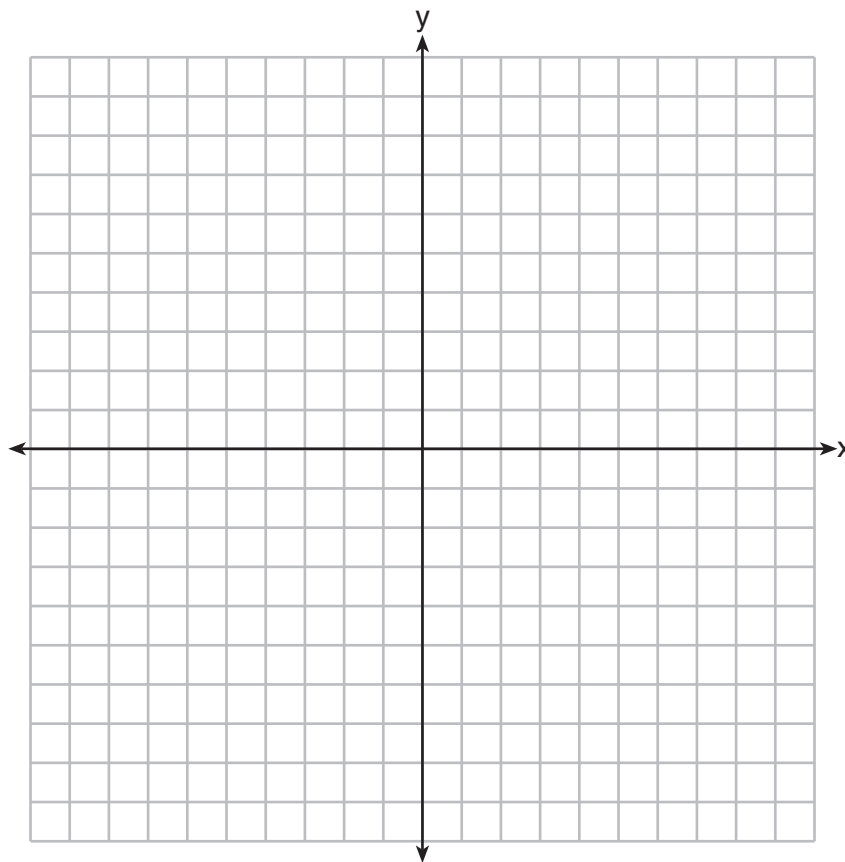
30 Triangle ABC and triangle DEF are drawn below.



If $\overline{AB} \cong \overline{DE}$, $\overline{AC} \cong \overline{DF}$, and $\angle A \cong \angle D$, write a sequence of transformations that maps triangle ABC onto triangle DEF .

- 31** Line n is represented by the equation $3x + 4y = 20$. Determine and state the equation of line p , the image of line n , after a dilation of scale factor $\frac{1}{3}$ centered at the point $(4,2)$.
[The use of the set of axes below is optional.]

Explain your answer.



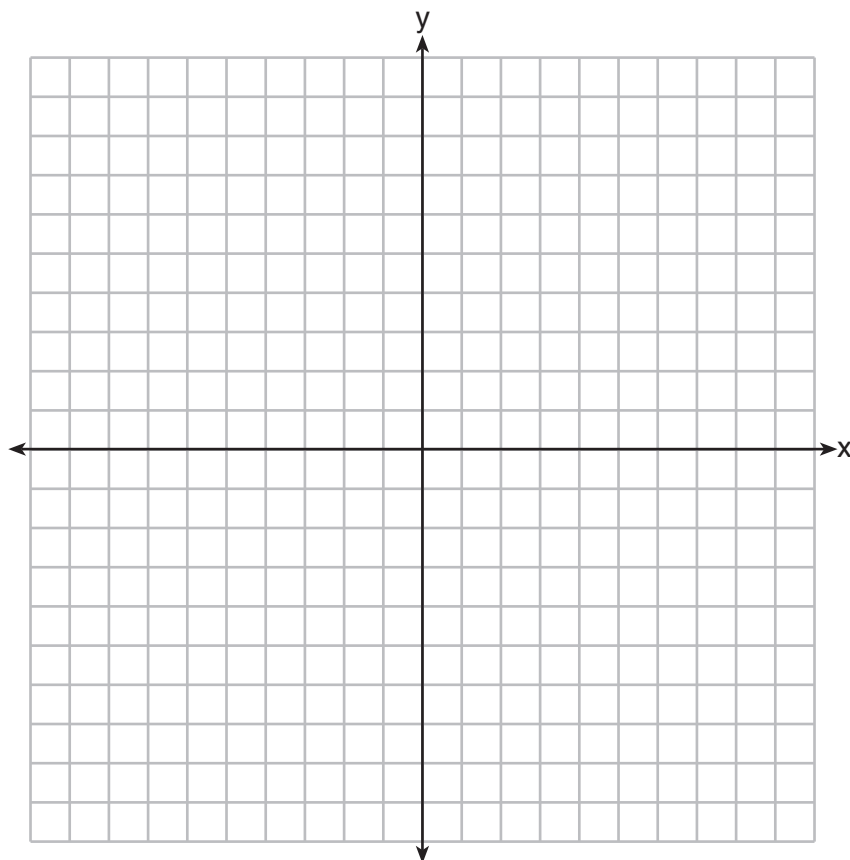
Part III

Answer all 3 questions in this part. Each correct answer will receive 4 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [12]

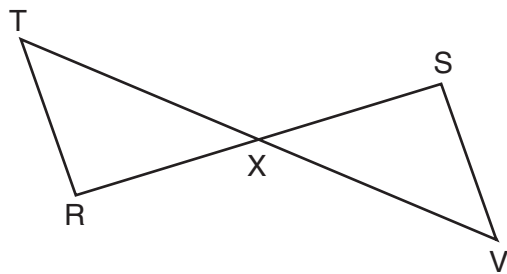
- 32 Triangle ABC has vertices at $A(-5,2)$, $B(-4,7)$, and $C(-2,7)$, and triangle DEF has vertices at $D(3,2)$, $E(2,7)$, and $F(0,7)$. Graph and label $\triangle ABC$ and $\triangle DEF$ on the set of axes below.

Determine and state the single transformation where $\triangle DEF$ is the image of $\triangle ABC$.

Use your transformation to explain why $\triangle ABC \cong \triangle DEF$.

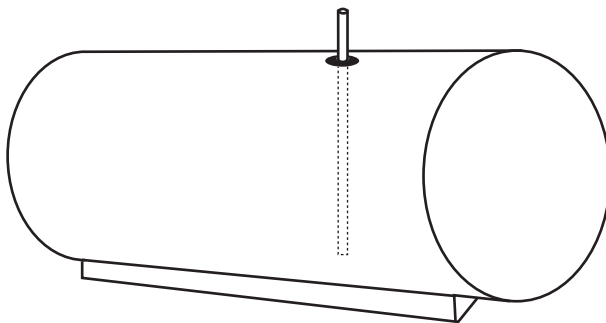


33 Given: $\overline{RS}$ and $\overline{TV}$ bisect each other at point X
 $\overline{TR}$ and $\overline{SV}$ are drawn



Prove: $\overline{TR} \parallel \overline{SV}$

- 34 A gas station has a cylindrical fueling tank that holds the gasoline for its pumps, as modeled below. The tank holds a maximum of 20,000 gallons of gasoline and has a length of 34.5 feet.



A metal pole is used to measure how much gas is in the tank. To the *nearest tenth of a foot*, how long does the pole need to be in order to reach the bottom of the tank and still extend one foot outside the tank? Justify your answer. [$1 \text{ ft}^3 = 7.48 \text{ gallons}$]

Part IV

Answer the 2 questions in this part. Each correct answer will receive 6 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [12]

35 Quadrilateral $PQRS$ has vertices $P(-2,3)$, $Q(3,8)$, $R(4,1)$, and $S(-1,-4)$.

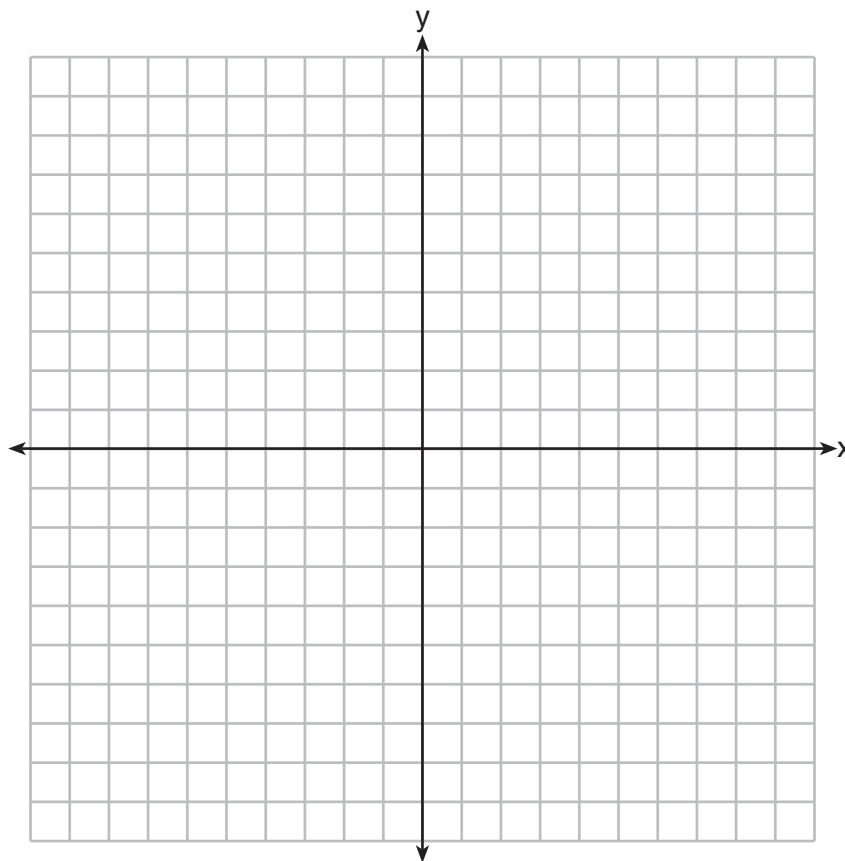
Prove that $PQRS$ is a rhombus.

[The use of the set of axes on the next page is optional.]

Question 35 is continued on the next page.

Question 35 continued.

Prove that $PQRS$ is *not* a square.
[The use of the set of axes below is optional.]



36 Freda, who is training to use a radar system, detects an airplane flying at a constant speed and heading in a straight line to pass directly over her location. She sees the airplane at an angle of elevation of 15° and notes that it is maintaining a constant altitude of 6250 feet. One minute later, she sees the airplane at an angle of elevation of 52° . How far has the airplane traveled, to the nearest foot?

Determine and state the speed of the airplane, to the nearest mile per hour.

If the student’s responses for the multiple-choice questions are being hand scored prior to being scanned, the scorer must be careful not to make any marks on the answer sheet except to record the scores in the designated score boxes. Marks elsewhere on the answer sheet will interfere with the accuracy of the scanning.

Part I

Allow a total of 48 credits, 2 credits for each of the following. Allow credit if the student has written the correct answer instead of the numeral 1, 2, 3, or 4.

(1) 2	(9) 2	(17) 4
(2) 3	(10) 2	(18) 1
(3) 3	(11) 4	(19) 2
(4) 4	(12) 1	(20) 2
(5) 4	(13) 1	(21) 4
(6) 3	(14) *	(22) **
(7) 1	(15) 2	(23) 3
(8) 2	(16) 1	(24) **

*** Question 14** — When scoring this question, either choice 1 or choice 3 should be awarded credit.

****Questions 22 and 24** — When scoring these questions, all students should be awarded credit regardless of the answer, if any, they record on the answer sheet for this question.

Updated information regarding the rating of this examination may be posted on the New York State Education Department’s web site during the rating period. Check this web site at: <http://www.p12.nysed.gov/assessment/> and select the link “Scoring Information” for any recently posted information regarding this examination. This site should be checked before the rating process for this examination begins and several times throughout the Regents Examination period.

The Department is providing supplemental scoring guidance, the “Model Response Set,” for the Regents Examination in Geometry (Common Core). This guidance is intended to be part of the scorer training. Schools should use the Model Response Set along with the rubrics in the Scoring Key and Rating Guide to help guide scoring of student work. While not reflective of all scenarios, the Model Response Set illustrates how less common student responses to constructed-response questions may be scored. The Model Response Set will be available on the Department’s web site at: <http://www.nysedregents.org/geometrycc/>.

Part II

For each question, use the specific criteria to award a maximum of 2 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

- (25) [2] A correct construction is drawn showing all appropriate arcs.
- [1] An appropriate construction is drawn showing all appropriate arcs, but an altitude is drawn from a vertex other than J .
- or**
- [1] An appropriate construction is drawn showing all appropriate arcs, but the altitude is missing or incorrect.
- [0] A drawing that is not an appropriate construction is shown.
- or**
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
- (26) [2] 2.25π or an equivalent area in terms of pi is written, and appropriate work is shown.
- [1] Appropriate work is shown, but one computational error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- or**
- [1] 2.25π , but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
- (27) [2] A complete and correct explanation is written.
- [1] An appropriate explanation is written, but one conceptual error is made.
- or**
- [1] An incomplete or partially correct explanation is written.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (28) [2] 0.6, and correct work is shown.
- [1] Appropriate work is shown, but one computational or rounding error is made.
- or***
- [1] Appropriate work is shown, but one conceptual error is made.
- or***
- [1] 0.6, but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
- (29) [2] A complete and correct explanation is written.
- [1] An explanation that contains one conceptual error is written.
- or***
- [1] A correct explanation of why one pair of angles is congruent is written.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
- (30) [2] A correct sequence of transformations is written.
- [1] An appropriate sequence of transformations is written, but one conceptual error is made.
- or***
- [1] An appropriate sequence of transformations is written, but it is incomplete.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(31) [2] $y = -\frac{3}{4}x + 5$ or an equivalent equation is written, and a correct explanation is written.

[1] Appropriate work is shown, but one computational or graphing error is made. A correct explanation is written.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] $y = -\frac{3}{4}x + 5$ or an equivalent equation is written, but the explanation is incomplete or incorrect.

[0] The equation $3x + 4y = 20$ or an equivalent equation is written, but the explanation is missing.

or

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

Part III

For each question, use the specific criteria to award a maximum of 4 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(32) [4] Triangles ABC and DEF are graphed and labeled correctly, a reflection over the correct line is stated, and a correct explanation is written.

[3] Appropriate work is shown, but one or more graphing or labeling errors are made. An appropriate line of reflection is stated, and an appropriate explanation is written.

or

[3] Appropriate work is shown, but the line of reflection is missing or incorrect. An appropriate explanation is written.

[2] Appropriate work is shown, but one or more graphing or labeling errors are made. An appropriate line of reflection is stated, but an incomplete or partially correct explanation is written.

or

[2] Appropriate work is shown to graph and label both triangles, and a correct line of reflection is stated. No further correct work is shown.

[1] Appropriate work is shown to graph and label both triangles, but no further correct work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (33) [4] A complete and correct proof that includes a conclusion is written.
- [3] A proof is written that demonstrates a thorough understanding of the method of proof and contains no conceptual errors, but one statement and/or reason is missing or incorrect.
- [2] A proof is written that demonstrates a good understanding of the method of proof and contains no conceptual errors, but two statements and/or reasons are missing or incorrect.
- or***
- [2] A proof is written that demonstrates a good understanding of the method of proof, but one conceptual error is made.
- or***
- [2] A proof is written that shows $\triangle TXR \cong \triangle VXS$, but no further correct work is shown.
- [1] Only one correct statement and reason are written.
- [0] The “given” and/or the “prove” statements are written, but no further correct relevant statements are written.
- or***
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (34) [4] 10.9, and correct work is shown.
- [3] Appropriate work is shown, but one computational or rounding error is made.
- or***
- [3] Correct work is shown to find the radius of the cylinder, but no further correct work is shown.
- [2] Appropriate work is shown, but one conceptual error is made.
- or***
- [2] Appropriate work is shown, but two or more computational or rounding errors are made.
- [1] Appropriate work is shown, but one conceptual error and one computational or rounding error are made.
- or***
- [1] Appropriate work is shown to find the number of cubic feet in the tank, but no further correct work is shown.
- or***
- [1] 10.9, but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

Part IV

For each question, use the specific criteria to award a maximum of 6 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(35) [6] A complete and correct proof that includes concluding statements that $PQRS$ is a rhombus and $PQRS$ is not a square is written.

[5] Appropriate work is shown, but one computational or graphing error is made. Appropriate concluding statements are written.

or

[5] Appropriate work is shown to prove $PQRS$ is a rhombus, and work is shown to prove $PQRS$ is not a square. One concluding statement is missing or incorrect.

[4] Appropriate work is shown, but two or more computational or graphing errors are made. Appropriate concluding statements are written.

or

[4] Appropriate work is shown, but one conceptual error is made. Appropriate concluding statements are written.

or

[4] Appropriate work is shown to prove $PQRS$ is a rhombus and a concluding statement is written. No further correct work is shown.

[3] Appropriate work is shown, but one conceptual error and one computational or graphing error are made. Appropriate concluding statements are written.

or

[3] Appropriate work is shown to prove $PQRS$ is a parallelogram and a concluding statement is written. No further correct work is shown.

[2] Appropriate work is shown, but two conceptual errors are made. Appropriate concluding statements are written.

or

[2] Appropriate work is shown, but one conceptual error and two or more computational or graphing errors are made. Appropriate concluding statements are written.

or

[2] Appropriate work is shown to prove two pairs of opposite sides are parallel. No further correct work is shown.

or

[2] Appropriate work is shown to find the lengths of all four sides. No further correct work is shown.

or

[2] Appropriate work is shown to prove the diagonals are perpendicular bisectors of each other. No further correct work is shown.

or

[2] Appropriate work is shown to prove $PQRS$ is not a square and a concluding statement is written. No further correct work is shown.

[1] Appropriate work is shown, but two conceptual errors and one computational or graphing error are made. Appropriate concluding statements are written.

or

[1] Appropriate work is shown to find the slopes of all four sides. No further correct work is shown.

or

[1] Appropriate work is shown to find the slopes and lengths of one pair of opposite sides. No further correct work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (36) [6] 18,442 and 210, and correct work is shown.
- [5] Appropriate work is shown, but one computational or rounding error is made.
- or**
- [5] Correct work is shown to find 18,442, and the speed of the airplane in miles per minute or feet per hour, but no further correct work is shown.
- [4] Appropriate work is shown, but two computational or rounding errors are made.
- or**
- [4] Appropriate work is shown, but one conceptual error is made.
- or**
- [4] Correct work is shown to find the distance the airplane has traveled, 18,442, but no further correct work is shown.
- [3] Appropriate work is shown, but three or more computational or rounding errors are made.
- or**
- [3] Appropriate work is shown, but one conceptual error and one computational or rounding error are made.
- [2] Two correct trigonometric equations are written to determine how far the airplane has traveled, but no further correct work is shown.
- or**
- [2] Appropriate work is shown, but one conceptual error and two or more computational or rounding errors are made.
- or**
- [2] Appropriate work is shown, but two conceptual errors are made.
- [1] Appropriate work is shown, but two conceptual errors and one computational or rounding error are made.
- or**
- [1] 18,442 and 210, but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

**Map to the Common Core Learning Standards
Geometry (Common Core)
June 2017**

Question	Type	Credits	Cluster
1	Multiple Choice	2	G-CO.B
2	Multiple Choice	2	G-GPE.B
3	Multiple Choice	2	G-SRT.C
4	Multiple Choice	2	G-C.A
5	Multiple Choice	2	G-SRT.B
6	Multiple Choice	2	G-SRT.A
7	Multiple Choice	2	G-CO.A
8	Multiple Choice	2	G-C.A
9	Multiple Choice	2	G-CO.C
10	Multiple Choice	2	G-SRT.B
11	Multiple Choice	2	G-CO.C
12	Multiple Choice	2	G-GPE.A
13	Multiple Choice	2	G-SRT.C
14	Multiple Choice	2	G-SRT.A
15	Multiple Choice	2	G-GPE.B
16	Multiple Choice	2	G-GMD.A
17	Multiple Choice	2	G-CO.C
18	Multiple Choice	2	G-GMD.B
19	Multiple Choice	2	G-GPE.B
20	Multiple Choice	2	G-CO.C
21	Multiple Choice	2	G-SRT.C
22	Multiple Choice	2	G-CO.B
23	Multiple Choice	2	G-MG.A
24	Multiple Choice	2	G-SRT.B
25	Constructed Response	2	G-CO.D
26	Constructed Response	2	G-C.B
27	Constructed Response	2	G-GMD.A
28	Constructed Response	2	G-MG.A
29	Constructed Response	2	G-SRT.B
30	Constructed Response	2	G-CO.A
31	Constructed Response	2	G-SRT.A
32	Constructed Response	4	G-CO.B
33	Constructed Response	4	G-CO.C
34	Constructed Response	4	G-MG.A
35	Constructed Response	6	G-GPE.B
36	Constructed Response	6	G-SRT.C